# Yunhan Qiao

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Ph.D. Candidate, Computer Science, Oregon State University

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## Research Interests

My research sits at the intersection of empirical software engineering, code comprehension, and human–AI collaboration. I study how students and experienced developers work with GenAI coding assistants and agents (e.g., GitHub Copilot, Cursor) in realistic *brownfield* codebases, and whether faster, more correct work also means deeper understanding. I combine controlled experiments, screen-recorded behavioral analysis, prompt coding, EEG, and interviews to inform how computing education teaches and assesses GenAI-assisted development and how future AI coding agents and IDEs should be designed.

## Education

**Oregon State University** — Corvallis, OR, United States
*Ph.D. in Computer Science, School of EECS* — *Sep 2023 – Jun 2027 (expected)*
**Advisor**: Dr. Christopher Hundhausen at the SEAL Lab
**Research Areas**: Empirical Software Engineering, Code Comprehension, Computing Education, Human–AI Interaction

**Arizona State University** — Tempe, AZ, United States
*M.S. in Software Engineering* — *Aug 2021 – May 2023*
**Advisor**: Dr. Robert LiKamWa
**Thesis**: Networked System for Volumetric Athletic Coaching in Augmented Reality

**Guangxi University of Science and Technology** — Liuzhou, China
*B.E. in Software Engineering* — *Sep 2016 – Jun 2020*

**Southern Cross University** — Gold Coast, Australia
*B.E. in Information Technology* — *Sep 2017 – Jun 2020*

## Publications (Google Scholar Page)

### Peer-Reviewed Journal Articles

**[J2]** Tariq, A., Conrad, P., Hundhausen, C., Chandrasekaran, S., **Qiao, Y.**, & Haque, S. (2026). Using SlackBot-prompted reflections to promote contribution equality in team software development projects. *ACM Transactions on Computing Education.* https://doi.org/10.1145/3857221

**[J1]** **Qiao, Y.**, Shihab, M. I. H., & Hundhausen, C. (2026). A systematic literature review of the use of GenAI assistants for code comprehension: Implications for computing education research and practice. *ACM Transactions on Computing Education, 26*(2), Article 24, 1–33. https://doi.org/10.1145/3785366
*Selected by the TOCE editorial board for presentation at SIGCSE TS 2027.*

### Peer-Reviewed Conference Proceedings

**[C2]** Motwani, M., Kulkarni, A., **Qiao, Y.**, Davis, M., & Chen, Z. (2025). LLM-guided differential fuzzing for detecting platform-specific bugs in scientific applications. *2025 IEEE Conference on AI x Software Engineering (AIxSE)*, 59–66. https://doi.org/10.1109/AIxSE64906.2025.00015

**[C1]** Shihab, M. I. H., Hundhausen, C., Tariq, A., Haque, S., **Qiao, Y.**, & Mulanda, B. W. (2025). The effects of GitHub Copilot on computing students' programming effectiveness, efficiency, and processes in brownfield coding tasks. *Proceedings of the 2025 ACM Conference on International Computing Education Research (ICER '25), 1*, 407–420. https://doi.org/10.1145/3702652.3744219

### Under Review

**[R2]** **Qiao, Y.**, Haque, S., & Hundhausen, C. (2026). Beyond the prompt: Linking what developers ask, do, and understand with coding agents. Under review at *IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER 2027).* https://arxiv.org/abs/2609.33712

**[R1]** **Qiao, Y.**, Shihab, M. I. H., Haque, S., & Hundhausen, C. (2025). Faster but not wiser: GitHub Copilot decouples programming performance from code comprehension in brownfield tasks. Under review at *ACM Transactions on Computing Education.* https://arxiv.org/abs/2511.02922

### Under Submission

**[S1]** Schutz, O., **Qiao, Y.**, Moussaoui, F. A., Choudhuri, R., & Sarma, A. (2026). Heart and mind in the loop: Why we vibe when we work with AI. Submitted to *International Conference on Cooperative and Human Aspects of Software Engineering (CHASE 2027).*

## Teaching Experience

**Graduate Teaching Assistant, School of EECS** — Oregon State University
*Undergraduate software engineering courses* — *2024 – current*

**Courses**: CS 362 Software Engineering II; CS 461 Senior Software Engineering Project; CS 467 Online Capstone Project; CS 492 Mobile Software Development

## Professional Experience

**Software Engineering Intern, Mocha Software** Nanjing, China
*Software development* *Aug 2020 – Nov 2020*
Developed a conference system for China Mobile Migu Culture & Technology Group with Spring, Spring MVC, and MyBatis, using Ajax to cut page rendering time from 1 s to 0.5 s and dynamic SQL with Redis-backed SSO sessions to reduce search load.

## Awards & Scholarships

Oregon State University Scholarly Presentation Award *2026*
Oregon State University EECS Graduate Scholarship *2023 – current*
Guangxi University of Science and Technology Merit Award *2018*

## Academic Service

**Program Committee**
ACM Technical Symposium on Computer Science Education (SIGCSE TS), 2027
IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), 2027

**Reviewer**
ACM Transactions on Computing Education (TOCE), 2027

## Skills & Coursework

**Research Methods**: Qualitative Analysis, Quantitative Analysis, Interviews, Survey Design
**Programming**: Python, Java, C, JavaScript, HTML, CSS, SQL, Bash
**Frameworks & Tools**: React, Express, Flask, MongoDB, PostgreSQL, Git, AWS, Docker, LaTeX, Figma; Claude Code, GitHub Copilot, Cursor, OpenAI Codex
**Coursework**: Machine Learning; NLP with Deep Learning; Software Engineering for ML; Program Analysis; Algorithms & Data Structures; Human–Computer Interaction; Field Studies in SE & HCI; Methods of Data Analysis

## Languages

English (Fluent), Mandarin Chinese (Native)